

PRINT Your Name: _____

PRINT Your Student ID: _____

PRINT Student name to your left: _____

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You have 170 minutes. There are 7 questions of varying credit. (100 points total)

Question:	1	2	3	4	5	6	7	Total
Points:	17	16	9	14	16	18	10	100

We reserve the right to deduct points for failing to follow the marking directions below:

For questions with **circular bubbles**, you may select only one choice.

☐ Unselected option (Completely unfilled)

☐ Don't do this (it will be graded as incorrect!)

☒ Only one selected option (completely filled)

For questions with **square checkboxes**, you may select one or more choices.

☐ You can select

☐ multiple squares

☒ Don't do this (it will be graded as incorrect!)

Anything you write outside the answer boxes or you ~~cross-out~~ will not be graded. If you write multiple answers, your answer is ambiguous, or the bubble/checkbox is not entirely filled in, we will count the lowest-scoring interpretation.

Read the honor code below and sign your name.

By signing below, I affirm that all work on this exam is my own work. I have not referenced any disallowed materials, nor collaborated with anyone else on this exam. I understand that if I cheat on the exam, I may face the penalty of an "F" grade and a referral to the Center for Student Conduct.

SIGN your name: _____

Q1 Potpourri**(17 points)**

Q1.1 (1 point) Fill in the blank with the best option: Running BFS (breadth-first) tree search and running BFS graph search on the same graph will _____ return paths of different length.

- ☐ (A) Always ☐ (B) Sometimes ☐ (C) Never

For Q1.2 to Q1.3, let h_1 and h_2 be admissible heuristics.

Q1.2 (1 point) True or false: A* tree search using h_1 may find a lower-cost solution than A* tree search using h_2 .

- ☐ (A) True ☐ (B) False

Q1.3 (1 point) Let h_3 be a heuristic such that $h_3(s) = \min(h_1(s), h_2(s))$ for all states s .

True or false: h_3 is admissible.

- ☐ (A) True ☐ (B) False

Q1.4 (1 point) Fill in the blank with the best option: Minimax with alpha-beta pruning will _____ return a different best action than minimax without alpha-beta pruning if the tree is unbalanced.

- ☐ (A) Always ☐ (B) Sometimes ☐ (C) Never

Q1.5 (1 point) Fill in the blank with the best option: If we change all the minimizer nodes in a minimax tree to chance nodes that pick actions uniformly at random, the value at the root will _____ increase.

- ☐ (A) Always ☐ (B) Sometimes ☐ (C) Never

For Q1.6 to Q1.7, consider the Gridworld below. In states A and E , the only action available is “Exit”, which earns the +100 or +1 reward, respectively. In states B , C , and D , the actions are “Left” and “Right” which are successful 100% of the time. There is no reward other than the +100 or +1.

+100				+1
A	B	C	D	E

Q1.6 (2 points) When running value iteration on this Gridworld, what is the smallest k for which $V_k(s) = V_{k+1}(s)$ for all states $s \in \{A, B, C, D, E\}$?

Assume that the discount factor is $\gamma = 1$ and that $V_0(s) = 0$ for all states $s \in \{A, B, C, D, E\}$.

- ☐ (A) 0 ☐ (B) 1 ☐ (C) 2 ☐ (D) 3 ☐ (E) 4 ☐ (F) 5

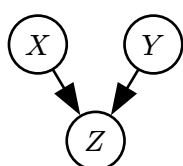
Q1.7 (2 points) For what discount factor are the actions “Left” and “Right” equally good when in state D ? Your answer should be a number.

(Question 1 continued...)

Q1.8 (1 point) The general Bellman update $V'(s) = \max_a \sum_{s'} T(s, a, s') [R(s, a, s') + \gamma V(s)]$ requires $O(SA)$ time to compute for one state.

Now, suppose the MDP is deterministic, with the probabilistic model $T(s, a, s')$ replaced by the function $\text{Result}(s, a)$. What is the big-O runtime now to compute the Bellman update for one state? Answer in big-O notation.

Q1.9 (1 point) Andrew is running Gibbs sampling on the following Bayes Net where each variable can take on the value 0 or 1. The probability table for Z is provided below (note that the table is equivalent to saying that $Z = X \text{ XOR } Y$):



X	Y	Z	$P(Z X, Y)$
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

The current state of the Gibbs sampling process is $X = 0, Y = 1, Z = 1$.

We choose to re-sample X . What is the probability that X will be re-sampled as 0?

☐ (A) 0

☐ (C) 1

☐ (B) 0.5

☐ (D) None of the above

Q1.10 (1 point) True or false: In a Hidden Markov Model (HMM), particle filtering generally gives more accurate values for $P(X_t | e_{1:t})$ than the forward algorithm.

☐ (A) True

☐ (B) False

Q1.11 (2 points) Which of the following represent *advantages* of transformer models covered in class *over* trigram models?

☐ (A) Allow adjacent-word correlations to be captured

☐ (B) Allow distant-word correlations to be captured

☐ (C) Allow words that are more frequent in the training data to be given higher probabilities

☐ (D) Can assign nonzero probabilities to sentences never seen in the training data

☐ (E) None of the above

(Question 1 continued...)

Q1.12 (1 point) Isabella wants to create an AI assistant for students in CS188. She first pre-trains a language model on transcripts of CS188 lectures. When she evaluates it on a test set of CS188 lectures, she finds that the model achieves high accuracy on predicting the next token.

Which of the following is the best next step for Isabella to take?

- Ⓐ Use a tokenizer that is trained on lectures from more courses, not just CS188
- Ⓑ Fine-tune the model on helpful human conversations
- Ⓒ Train the model on CS188 lectures using self-supervised learning for additional epochs
- Ⓓ Increase the size of the weight matrices used in the attention layers

Q1.13 (1 point) Kanav is using a word bigram model. Select the operator to fill in the blanks that makes the following equation true under this model:

$$P(\text{"Pac-Man wins"}) = P(\text{"Pac-Man"} \mid \text{START}) \text{ ______ } P(\text{"wins"} \mid \text{"Pac-Man"}) \text{ ______ } P(\text{STOP} \mid \text{"wins"})$$

Ⓐ + Ⓑ − Ⓒ × Ⓓ ÷

Q1.14 (1 point) Which of the following is the most suitable method to prevent overfitting in a decision tree?

- Ⓐ Minimizing the information gain at the root node
- Ⓑ Maximizing the average entropy of the label distributions at the leaves of the tree
- Ⓒ Removing a random subset of data points from the training data
- Ⓓ Limiting the depth of the decision tree

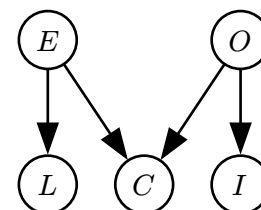
Q2 Bayes Nets: Printer Problems

(16 points)

It's the night before the CS188 final exam, and Niklas and Aly are printing copies of the exam. We use the Bayes Net to the right to model whether the printer will print a single exam correctly.

The variables in our Bayes Net are

- E : is the toner empty?
- L : is the empty toner light on?
- O : is the printer out of paper?
- I : is the paper indicator on?
- C : does the printer print the exam correctly?



Assume each variable is binary (e.g., E can take on values e or $\neg e$).

Q2.1 (2 points) Select all of the following variables that are guaranteed to be **independent** of L given E .

☐ A C

☐ C I

☐ B O

☐ D None of the above

Q2.2 (2 points) Select all of the following variables that are guaranteed to be **dependent** on I given C .

☐ A O

☐ C L

☐ B E

☐ D None of the above

Q2.3 (2 points) How many entries are there in total across all probability tables in the Bayes Net? *Ignore* entries whose values are determined by the sum-to-1 constraint. E.g., $P(X)$ for a three-valued variable X is just two entries because the third is determined by the first two.

Q2.4 (2 points) Niklas wants to calculate $P(C \mid i)$ using variable elimination and decides to eliminate the variable O first. Which new factor f does this create?

☐ A $f(O)$

☐ D $f(O, C)$

☐ G $f(E, C, i)$

☐ I $f(L, E, C, i)$

☐ B $f(C)$

☐ E $f(C, i)$

☐ H $f(O, E, C)$

☐ K $f(L, O, E, C, i)$

☐ C $f(E, C)$

☐ F $f(O, C, i)$

☐ I $f(O, E, C, i)$

☐ L None of the above

Q2.5 (1 point) Which of the following orders is a valid visitation order for applying prior sampling to this Bayes net?

☐ A L, E, C, O, I

☐ B C, E, I, L, O

☐ C E, L, O, C, I

☐ D None of the above

(Question 2 continued...)

Q2.6 (1 point) Fill in the blank with the best option: In the limit as the number of samples tends to infinity, if we run rejection sampling to estimate $P(E \mid l)$, we will ____ reject more samples than if we run rejection sampling to estimate $P(E \mid l, \neg c)$.

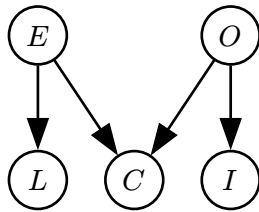
- Ⓐ Always
- Ⓑ Sometimes
- Ⓒ Never
- Ⓓ Cannot tell with the provided information

Q2.7 (2 points) For this part only, we find that there's an app that can tell us when there is a problem with the toner and/or paper. We want to include this in our Bayes Net.

Let the app variable be represented by A . A can take on one of two values:

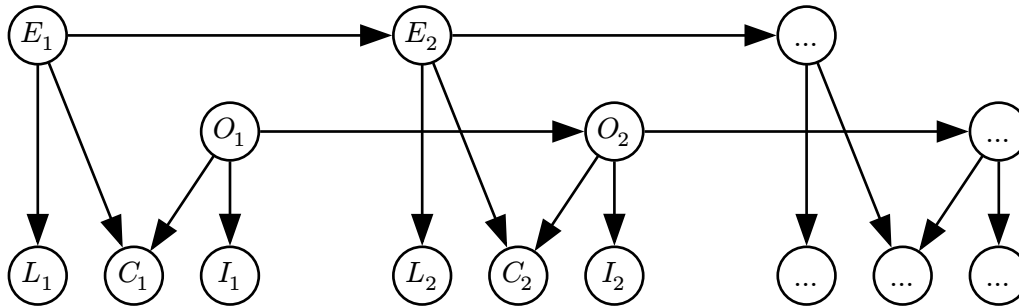
- a : signals that there is a problem (toner is empty and/or paper is out)
- $\neg a$: signals that there is not a problem (toner is not empty and paper is not out)

Add the appropriate nodes and edges to the Bayes Net below.



(Question 2 continued...)

For Q2.8 to Q2.9, we use the following dynamic Bayes Net to model printing out multiple different copies of the exam, where each time step represents printing out one copy. We show the first two time steps, and the pattern repeats after that:



Q2.8 (2 points) For this part only, we use likelihood weighting to compute $P(E_2 \mid L_2 = l, I_2 = \neg i)$.

We generate the sample:

$(E_1 = \neg e, L_1 = \neg l, O_1 = o, I_1 = i, C_1 = c, E_2 = e, L_2 = l, O_2 = \neg o, I_2 = \neg i, C_2 = \neg c)$

Select the set of probability distributions that we need to compute the *weight* of this sample (not the sample itself).

- ☐ A $P(L_t = l)$
☐ D $P(I_t = i)$
☐ G $P(E_1)$
- ☐ B $P(L_t = l \mid E_t = e)$
☐ E $P(I_t = i \mid O_t = o)$
☐ H $P(E_{t+1} \mid E_t = \neg e)$
- ☐ C $P(L_t = l \mid E_t = \neg e)$
☐ F $P(I_t = i \mid O_t = \neg o)$
☐ I None of the above

Q2.9 (2 points) Consider the following probability table for $P(E_{t+1} \mid E_t)$:

E_t	E_{t+1}	$P(E_{t+1} \mid E_t)$
e	e	1
e	$\neg e$	0
$\neg e$	e	0.01
$\neg e$	$\neg e$	0.99

As the time step t approaches ∞ , what does $P(E_t = e)$ approach?

- ☐ A 0
 ☐ C 0.5
 ☐ E 1
- ☐ B 0.01
 ☐ D 0.99
 ☐ F None of the above

Q3 Please Don't Click on the Mines!**(9 points)**

Advika is playing a game on the following 3 x 3 board:

2	x_2	x_3
2	x_5	x_6
x_7	x_8	1

Some squares contain mines, and Advika has to avoid clicking on them. The squares labeled x_2, x_3, x_5, x_6, x_7 and x_8 have not yet been clicked on, and their contents are unknown.

Advika has already clicked on the squares marked 2, 2, and 1. (Thank goodness she is still alive!) Clicking on a square results in either death (if there is a mine) or reveals the number of mines in horizontally, vertically, and diagonally adjacent squares. For example, the 1 in the lower-right corner of the board means that there is 1 mine total under the three adjacent squares labeled x_5, x_6 , and x_8 .

To help her decide where it is safe to click next, Advika formulates a CSP. The variables x_i can take on values in $\{0, 1\}$ where $x_i = 1$ means there is a mine in that square, and $x_i = 0$ means there is not. (Note that the CSP applies only to this turn; after she clicks, Advika will need to update the CSP accordingly.) The constraints in the CSP arise from the mine-count information revealed by previous clicks.

Q3.1 (1 point) Which of the following constraints correctly reflect the current situation? Select all that apply.

☐ **A** $x_2 + x_5 = 2$

☐ **C** $x_6 + x_8 = 1$

☐ **B** $x_2 + x_5 + x_7 + x_8 = 1$

☐ **D** None of the above

Q3.2 (2 points) Advika is running backtracking search with forward checking. Her search assigns variables in the order $x_2, x_5, x_3, x_6, x_7, x_8$. Values are assigned in lexicographic order (0 then 1).

She assigns $x_2 = 0$. What happens next?

- ☐ **A** Advika assigns $x_5 = 0$ because x_5 is the next variable and 0 is the first value to try.
- ☐ **B** Advika assigns $x_5 = 0$ because the arc from $x_5 \rightarrow x_2$ is consistent if $x_5 = 0$.
- ☐ **C** Advika backtracks because there are no valid assignments for x_5 .
- ☐ **D** Advika backtracks because the depth-first search will run into a cycle.
- ☐ **E** None of the above

(Question 3 continued...)

Q3.3 (2 points) For this subpart only, let us consider general instances of this family of CSPs and suppose we add a new constraint: $\sum_i x_i \leq N$ where N is the maximum number of mines allowed on the grid. Which of the following are true?

- ☐ A The number of satisfying solutions to the CSP may increase.
- ☐ B The number of satisfying solutions to the CSP may stay the same.
- ☐ C The number of satisfying solutions to the CSP may decrease.
- ☐ D None of the above

Q3.4 (2 points) Backtracking search yields a solution to the CSP where $x_3 = 0$. Can Advika be certain that the corresponding square is safe?

- ☐ A Yes, because backtracking yields correct solutions to CSP problems.
- ☐ B Yes, because backtracking search works with higher-order constraints.
- ☐ C Yes, because of another reason not listed above
- ☐ D No, because backtracking does not always yield correct solutions to CSP problems.
- ☐ E No, because backtracking only works with binary constraints.
- ☐ F No, because of another reason not listed above.

Q3.5 (2 points) Advika notices that she can split this CSP into two disjoint CSPs that can be solved independently: CSP A and CSP B. She combines the solutions to CSP A and CSP B to generate a solution to the original CSP.

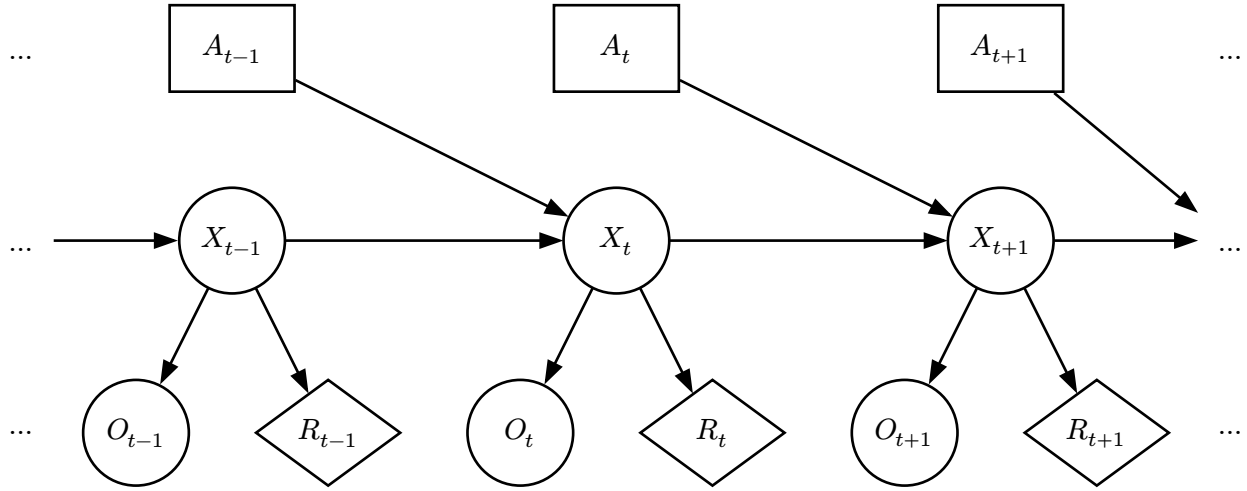
CSP A contains the x_5 variable. What are the other variables in CSP A?

- | | |
|----------------------------------|---|
| ☐ A x_2 | ☐ D x_7 |
| ☐ B x_3 | ☐ E x_8 |
| ☐ C x_6 | ☐ F None of the above |

Q4 Dynamic Decision Network

(14 points)

A dynamic decision network is dynamic Bayes net extended with action and reward nodes. In the example below, X_t is the state of the environment at time t , A_t is the action, R_t is the reward, and O_t is the observation. Note the indexing of variables: the edge goes from A_{t-1} to X_t . By convention, O_t is observed before A_t is chosen.



Suppose we want to maximize the sum of rewards $\sum_{t'=0}^T R_{t'}$ and we're currently at time step t (we've taken actions $A_{1:t-1}$ and observed $O_{1:t}$). Select the conditions under which each equation holds.

Q4.1 (1 point) $VPI(O_{t+1}|X_{t+1}) = 0$

- Ⓐ Always Ⓑ Sometimes Ⓒ Never

Q4.2 (1 point) $VPI(X_t, O_{t+1}) < VPI(X_t)$

- Ⓐ Always Ⓑ Sometimes Ⓒ Never

Q4.3 (1 point) $VPI(X_t, X_{t+1}) > VPI(X_t)$

- Ⓐ Always Ⓑ Sometimes Ⓒ Never

(Question 4 continued...)

Let the *belief state* $\mathbf{f}_t = P(X_t | o_{1:t}, a_{1:t-1})$ be a distribution representing the agent's belief over the hidden state at timestep t given the observations up to this point as well as all previous actions.

Q4.4 (2 points) Which of the following elements are required for us to compute a belief update: computing $\mathbf{f}_t(X_t)$ from $\mathbf{f}_{t-1}(X_{t-1})$?

- ☐ A $P(X_t | X_{t-1}, a_{t-1})$
- ☐ B $P(X_t | X_{t-1}, a_t)$
- ☐ C $P(X_t | X_{t-1})$
- ☐ D $P(o_t | X_t)$
- ☐ E $P(X_t | o_t)$
- ☐ F None of the above

Q4.5 (2 points) An optimal action a_t at timestep t is an action that maximizes the expected sum $\mathbb{E}[\sum_{t'=t}^T r_{t'} | o_{1:t}, a_{1:t-1}]$ of future rewards given previous observations and actions. In our dynamic decision network, knowing just $\mathbf{f}_t$ is sufficient to take an optimal action at timestep t . Why is this?

- ☐ A It stores the most likely hidden state
- ☐ B The future observations are conditionally independent of future actions, given $\mathbf{f}_t$
- ☐ C The distribution over future trajectories is conditionally independent of previous actions and observations, given $\mathbf{f}_t$
- ☐ D The belief state $\mathbf{f}_t$ is computed recursively in a way that allows you to reconstruct all previous observations $o_{1:t}$ and actions $a_{1:t-1}$
- ☐ E The current observation o_t contains all information relevant to future rewards
- ☐ F None of the above

Q4.6 (2 points) Suppose we want to use our dynamic decision network to model a gridworld environment in which we receive a reward of +1 every time we take the $A=\text{MoveNorth}$ action. How do we have to change the graphical model to model this?

- ☐ A Add an edge from O_t to X_{t+1}
- ☐ B Add an edge from O_t to R_t
- ☐ C Add an edge from R_t to A_{t-1}
- ☐ D Add an edge from A_{t-1} to O_t
- ☐ E Add an edge from R_{t+1} to X_t
- ☐ F None of the above

Q4.7 (2 points) Suppose we want to use our dynamic decision network to model a gridworld environment where the $A=\text{TakeMeasurement}$ action reveals our current state. How do we have to change the graphical model to model this?

- ☐ A Add an edge from O_t to X_{t+1}
- ☐ B Add an edge from A_t to X_t
- ☐ C Add an edge from X_t to A_t
- ☐ D Add an edge from A_{t-1} to O_t
- ☐ E Add an edge from A_{t-1} to R_t
- ☐ F None of the above

(Question 4 continued...)

Q4.8 (2 points) In which of the following cases are our observations $o_{1:T}$ guaranteed to not affect the optimal actions $a_{1:T}$?

- ☐ A Observations are independent of the state
- ☐ B Observations are a deterministic function of the states
- ☐ C States are fully observable
- ☐ D State transitions are deterministic and known and x_0 is known
- ☐ E None of the above

Q4.9 (1 point) To estimate the belief state $f_t(X_t)$, we decide to use a particle filter. When is the observation o_t used in the particle filter update?

- ☐ A When propagating particles using the transition model
- ☐ B When weighting particles by likelihood
- ☐ C When sampling initial particles
- ☐ D Observations are not relevant to the particle filter update
- ☐ E None of the above

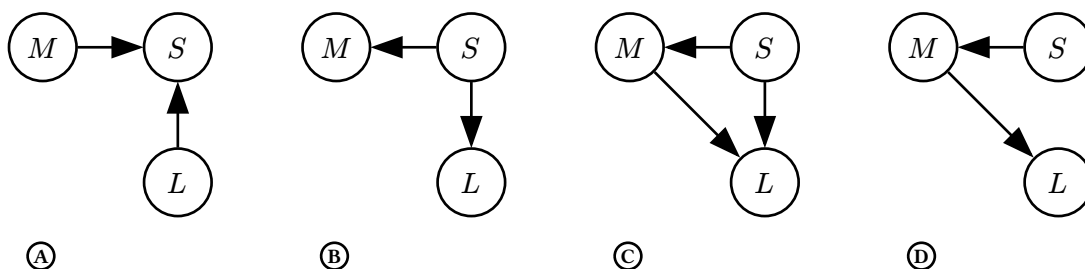
Q5 Naive Bass

(16 points)

You want to classify what species S a fish is. You recently learned that fish with upturned mouths can feed better at the water surface, and downturned mouths are adapted to feed at the bottom. You go to the aquarium and observe four fish, noting their mouth shape M (upturned or downturned), feeding location L (at the top or at the bottom of the tank) and species S (flatfish or bass), shown below. You build a Naive Bayes classifier using M and L , using the four fish as the training dataset D . Initially you use maximum likelihood estimation with no smoothing.

Fish ID	Mouth (M)	Feeding Location (L)	Species (S)
1	upturned	top	bass
2	downturned	bottom	flatfish
3	upturned	bottom	bass
4	downturned	top	bass

Q5.1 (1 point) Which of these Bayes nets corresponds to the Naive Bayes model of this classification problem?



Q5.2 (2 points) You spot a fish feeding at the bottom of its tank with an upturned mouth. According to your model, what is the probability that it is a bass?

- ☐ A 0 ☐ C $\frac{1}{3}$ ☐ E $\frac{2}{3}$ ☐ G 1
☐ B $\frac{1}{4}$ ☐ D $\frac{1}{2}$ ☐ F $\frac{3}{4}$ ☐ H None of the above

Q5.3 (2 points) What is the probability your model estimates a fish feeding at the bottom with a **downturned** mouth is a bass?

- ☐ A 0 ☐ C $\frac{1}{9}$ ☐ E $\frac{1}{4}$ ☐ G $\frac{1}{2}$ ☐ I $\frac{3}{4}$
☐ B $\frac{1}{10}$ ☐ D $\frac{1}{12}$ ☐ F $\frac{1}{3}$ ☐ H $\frac{2}{3}$ ☐ J None of the above

Q5.4 (3 points) Let P be the model created from the original training data D using maximum likelihood learning without smoothing. Denote the likelihood of data D under model P by $P(D)$. We now consider various ways to make a new Naive Bayes model P' , which is also unsmoothed unless otherwise specified. Select all of the following statements that are true, paying very careful attention to the exact statement of the inequality.

- ☐ **A** Collect data from *more fish* and combine them with the data in D to create dataset D' . Then train P' on D' to predict S from M and L . It is possible that $P'(D) < P(D)$.
- ☐ **B** Pick a new feature F and measure its value for each fish in D , to create dataset D' with features L , M , **and** F . Then train P' on D' to predict S from L , M , and F . It is possible that $P'(D') < P(D)$.
- ☐ **C** Define dataset D' as a copy of D with one of the features removed, then train P' on D' to predict S from the remaining feature alone. It is possible that $P'(D') < P(D)$.
- ☐ **D** P' has the same features as P and is trained on the same data D , but P' uses Laplace smoothing. It is possible that $P'(D) < P(D)$.
- ☐ **E** None of the above

(Question 5 continued...)

For the rest of the problem, your model uses data D and Laplace smoothing with strength k .

Q5.5 (1 point) What happens to your estimate of the probability that the next fish you spot is a flatfish, if you increase k from 0 to a positive value X ?

- Ⓐ Increases Ⓒ Stays the same Ⓔ None of the above
 Ⓑ Decreases Ⓓ Depends on X

Q5.6 (2 points) Which of the following are always true as k goes to infinity? Use P to denote the estimated probabilities under the Naive Bayes model.

- Ⓐ $P(\text{flatfish} \mid \text{downturned, bottom})$ approaches 1.
 Ⓑ $P(s, m, l)$ approaches $\frac{1}{2}$ for all values of s, m and l .
 Ⓒ $P(l|s)$ approaches $\frac{1}{2}$ for all values of l and s .
 Ⓓ The likelihood of the data in D under the model approaches 0.
 Ⓔ None of the above

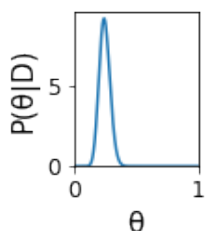
Q5.7 (3 points) Laplace smoothing with strength $k > 0$ assigns a value $P_L(S = s_j) = \frac{N_j + k}{\sum_i [N_i + k]}$ for the probability of the j th species after seeing N_j members of that species, where the sum is over all possible species. Before seeing any data, P_L assigns equal probability $\frac{k}{Mk} = \frac{1}{M}$ to each of M possible species.

Assume (as before) that the only species are bass and flatfish, and that you have not personally seen any fish yet. But you feel very confident, based on chatting with local fishing crews, that bass are about twice as common as flatfish. We can implement this idea by creating *weights* η for each species. Which one of the following formulae for $P_L(s_j)$ properly expresses your prior knowledge?

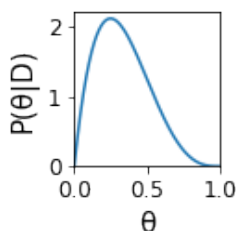
- Ⓐ $\frac{N_j + \eta_j \cdot k}{\sum_i [N_i + \eta_i \cdot k]}$, with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{3}$, small k
 Ⓑ $\frac{\eta_j \cdot N_j + k}{\sum_i [\eta_i \cdot N_i + k]}$ with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{3}$, small k
 Ⓒ $\frac{N_j + \eta_j \cdot k}{\sum_i [N_i + \eta_i \cdot k]}$, with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{2}$, small k
 Ⓓ $\frac{\eta_j \cdot N_j + k}{\sum_i [\eta_i \cdot N_i + k]}$ with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{2}$, small k
 Ⓔ $\frac{N_j + \eta_j \cdot k}{\sum_i [N_i + \eta_i \cdot k]}$, with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{3}$, large k
 Ⓕ $\frac{\eta_j \cdot N_j + k}{\sum_i [\eta_i \cdot N_i + k]}$ with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{3}$, large k
 Ⓖ $\frac{N_j + \eta_j \cdot k}{\sum_i [N_i + \eta_i \cdot k]}$, with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{2}$, large k
 Ⓗ $\frac{\eta_j \cdot N_j + k}{\sum_i [\eta_i \cdot N_i + k]}$ with $\eta_{\text{bass}} = 1, \eta_{\text{flatfish}} = \frac{1}{2}$, large k

(Question 5 continued...)

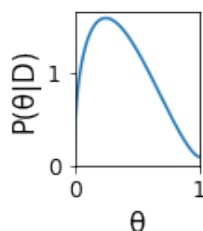
Q5.8 (2 points) Bayesian statisticians think of Laplace smoothing as a cheap heuristic approximation to real Bayesian learning! In lecture, we studied Bayesian learning in the context of a finite set of hypotheses (different kinds of bags of candy). But we can also apply it to a continuous family of hypotheses, such as the possible relative frequencies of bass and flatfish. That is, suppose $P(\text{flatfish}) = \theta$ and $P(\text{bass}) = 1 - \theta$. Then Bayesian learning can start with a prior probability $P(\theta)$ and can update that distribution to $P(\theta|D)$ as actual data D on bass and flatfish are observed. Suppose that you start with a uniform prior $P(\theta)$ over the range $[0,1]$ and then you observe one flatfish and three bass, as shown in the original dataset D . Which of the following best represents $P(\theta|D)$?



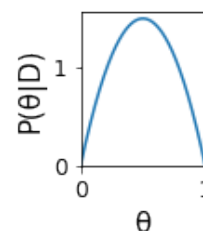
(A)



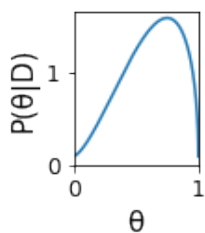
(B)



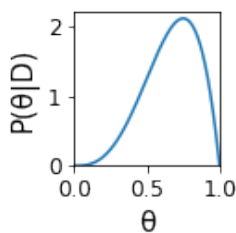
(C)



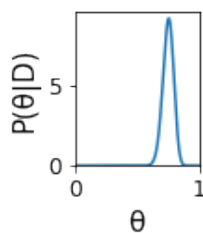
(D)



(E)



(F)



(G)

Q6 RL: Grid Racetrack

(18 points)

Oski is training a Reinforcement Learning agent to play the classic paper-and-pencil game Racetrack. The game is played on a 2D discrete grid. At any time step, the car's state is fully defined by its exact coordinates and velocity vector: $s = (x, y, v_x, v_y)$.

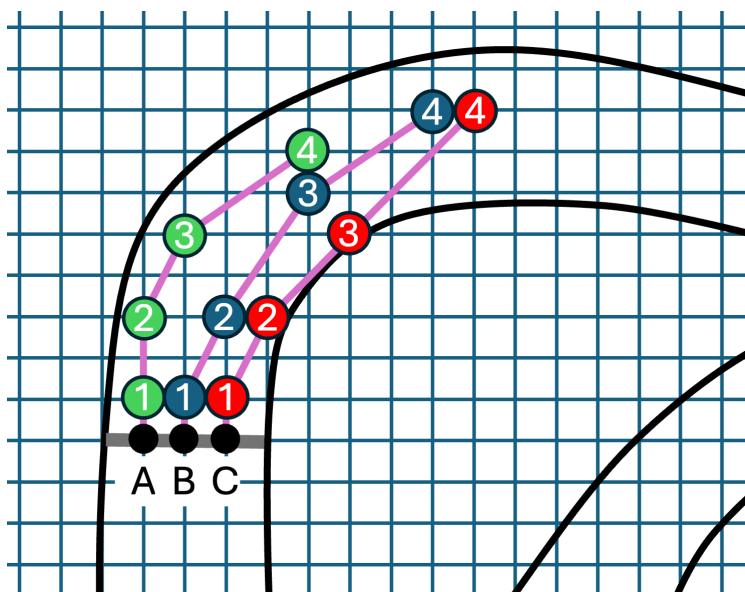
The possible actions are *accelerations* $a = (a_x, a_y)$, where both a_x and a_y can independently be $-1, 0$, or $+1$. The acceleration action updates the velocity ($v'_x = v_x + a_x$, $v'_y = v_y + a_y$) and then updates the location based on the new velocity ($x' = x + v'_x$, $y' = y + v'_y$).

The rewards are as follows:

- $+100$ for crossing the finish line in the correct direction.
- -100 for going off the track or crossing the finish line in the wrong direction. Both cases also lead to terminal states in which all actions have zero reward.
- -1 otherwise.

The discount factor is $\gamma = 0.5$ and the learning rate is $\alpha = 0.1$ unless otherwise specified.

The image below shows part of a racetrack and the first four timesteps of three different trajectories A, B, and C. Each starts on the finish line with velocity $(0,0)$ at time zero (the three black dots). The numbered circles show the positions at times 1, 2, 3, 4 for each trajectory.



Q6.1 (1 point) To make sure you have understood the rules, mark the trajectory that *violates* the rules, if any:

- ☐ A
 ☐ B
 ☐ C
 ☐ None of the above

Q6.2 (1 point) Which trajectory, if any, is guaranteed to go off the track?

- ☐ A
 ☐ B
 ☐ C
 ☐ None of the above

(Question 6 continued...)

Q6.3 (2 points) Consider the “Stanford policy” that stops in the middle of the track and then circumnavigates a unit square forever. What is the utility (sum of discounted rewards) of this policy?

- (A) -1.0 (C) -2.0 (E) $-\infty$
(B) -1.5 (D) -10.0 (F) None of the above

Q6.4 (3 points) Suppose the track fits inside an $N \times N$ square. Which of the following is the tightest upper bound on the size of the state space? [Hint: consider the maximum achievable x and y velocities.]

- (A) $O(N^2)$ (C) $O(N^3)$ (E) None of the above
(B) $O(N^{2.5})$ (D) $O(N^4)$

Q6.5 (3 points) Oski decides to solve the racetrack problem using Approximate Q-Learning with a linear value function $Q(s, a) = w_1 f_1(s, a) + w_2 f_2(s, a)$, where $w_1 = 5$ and $w_2 = -30$, and

- $f_1(s, a)$ = anticipated magnitude of speed after taking the action.
- $f_2(s, a) = 1$ if the proposed action goes outside the track and 0 otherwise.

Suppose that in the present state s , a proposed action a has $f_1(s, a) = 4$ and $f_2(s, a) = 1$.

The agent takes action a , receives a reward $r = -100$, and transitions to a terminal state s_0 where all actions have zero reward.

With a learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.5$, find the newly updated weight w_2 after the agent does a in s . (Hint: you may need to calculate the temporal difference error.)

- (A) -21 (C) -40 (E) -120
(B) -39 (D) -41 (F) None of the above

Q6.6 (2 points) Frustrated that the agent is driving too cautiously, Oski alters the reward function **for this subpart only**. He replaces the $r = -1$ step penalty with a positive reward equal to the updated speed: $R(s, a, s') = |v'_x| + |v'_y|$. Other rewards are unchanged.

Also, hoping to make winning more important, he sets γ to 0.99 **for this subpart only**.

Which of the following pathological behaviors is the agent most likely to learn?

- (A) The agent will safely drive to the finish line at the absolute maximum speed possible.
(B) The agent will avoid the finish line and instead drive in a safe, high-speed loop forever.
(C) The agent will immediately intentionally crash into a wall to terminate the episode and maximize expected utility.
(D) The agent will learn to remain perfectly still ($v_x = 0, v_y = 0$) to accumulate infinite zero-penalties.
(E) None of the above

(Question 6 continued...)

Q6.7 (2 points) Reverting to the standard reward setup and discount factor, Oski notices that his Q-learning agent has found a safe but very slow policy to get to the finish line. It sticks with the policy and never tries anything else. What might work to fix this?

- ☐ A Decreasing the learning rate α over time
- ☐ B Using an ε -greedy action selection strategy
- ☐ C Increasing the discount factor γ closer to 1
- ☐ D Giving a bonus reward proportional to $N(s, a)$, the number of times a has been tried in s
- ☐ E None of the above

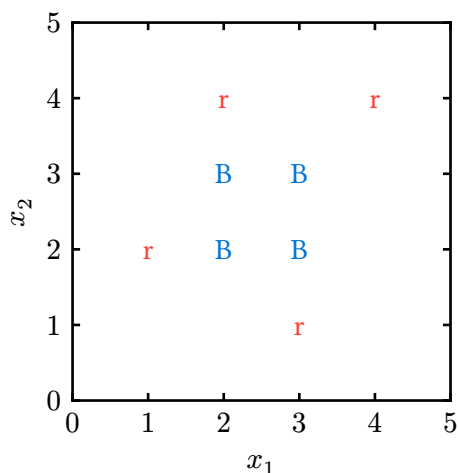
Q6.8 (4 points) Unhappy with the performance of his simple feature-based Q-learning agent, Oski trains a very large and deep neural network to represent the Q-function, with the inputs being the state (x, y, v_x, v_y) and the action (a_x, a_y) . After a few million trials, the Q-function is driving really well! Then he tries out the trained Q-function on a new track; much to his disappointment, it's a total failure. It soon veers across the track and crashes. Which of the following hypotheses and potential fixes are reasonable if Oski wants an agent that can drive well enough on any new track that comes along?

- ☐ A The network is too big and is memorizing rather than generalizing; a smaller network will work better.
- ☐ B Oski should train the Q-function on many different tracks and not just one; that will force it to generalize properly and work well for a new track.
The Q-function doesn't represent "how to drive well"; instead, it just approximates the Q-values for this particular track, which tell it what to do in each state. On a different track, it can be in the same state but a completely different action is needed.
- ☐ C The problem is with the input representation. The network needs an additional input indicating the identity number of the track; then multi-track training will work and the Q-function will generalize to new tracks.
- ☐ D The problem is with the input representation. The network will work better with an image of the track as an additional input; then multi-track training will work and the Q-function will generalize to new tracks.
- ☐ E None of the above

Q7 Five Nights with Machine Learning**(10 points)**

You are a security guard working the night shift at a local pizzeria. Each location (x_1, x_2) at the pizzeria has one animatronic (robot animal) present. Each animatronic is either a big bear (B) or a little rabbit (r).

For Q7.1 to Q7.3, you visit some locations and record the type of animatronic at each location. You graph the data in the following plot.



Q7.1 (1 point) It may be possible to use machine learning to predict what animal appears in each unexamined location. We can try training a single-layer perceptron, with input x_1, x_2 and output labels $+1$ for a bear and -1 for a rabbit. What happens on this data set, assuming a constant learning rate?

- Ⓐ The weights will always converge to a perfect classifier
- Ⓑ The weights will converge but with some misclassifications
- Ⓒ The weights will only converge if the learning rate is 0.5
- Ⓓ The weights are not guaranteed to converge
- Ⓔ None of the above

Q7.2 (1 point) While training, your perceptron incorrectly classifies location $(3, 2)$ as a rabbit. What happens to w_1 , the weight corresponding to feature 1? Assume that your learning rate is positive.

- Ⓐ Increases
- Ⓑ Decreases
- Ⓒ Stays the same

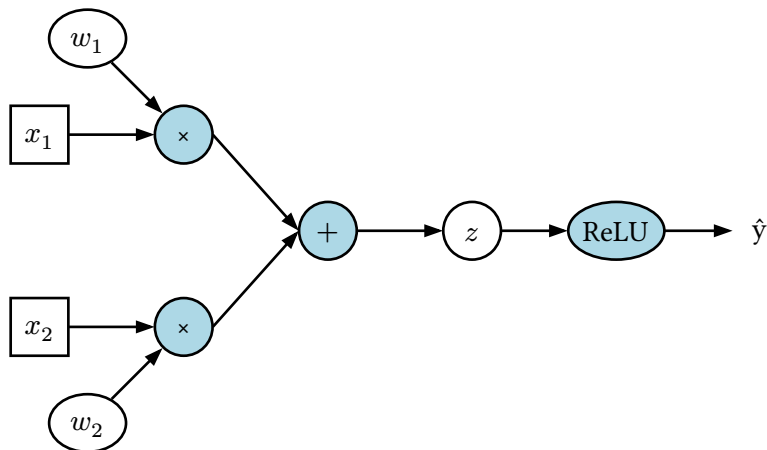
Q7.3 (2 points) You want to add a third feature, x_3 to help us better classify the data using a perceptron. Which of the following definitions for x_3 would allow a perceptron to classify the data perfectly?

- Ⓐ $x_3 = x_1 + x_2 + 2.5$
- Ⓑ $x_3 = x_1 - x_2$
- Ⓒ $x_3 = (x_1 - 2.5)^2 + (x_2 - 2.5)^2$
- Ⓓ $x_3 = (x_1 + x_2)^2 - 2.5$
- Ⓔ $x_3 = 2.5$
- Ⓕ None of the above

(Question 7 continued...)

For Q7.4 to Q7.6, you collect the heights of the animatronics at each location. You decide to build a neural network to predict the height of the animatronic at a location.

The neural network takes in two inputs, x_1 and x_2 , representing the location. The network outputs $\hat{y}$, which is the predicted height. The computation graph is shown below.



Note that $\hat{y} = \text{ReLU}(z)$. We use the loss function $L(y, \hat{y}) = \frac{1}{2}(y - \hat{y})^2$

Q7.4 (2 points) Let the weights be $w_1 = 1$ and $w_2 = -2$. Calculate $\hat{y}$ for the input $(x_1, x_2) = (1, 1)$.

Q7.5 (3 points) Now, we want to calculate $\frac{\partial L}{\partial w_1}$ during the backward pass. What partial derivatives are needed?

☐ A $\frac{\partial L}{\partial x_1}$

☐ F $\frac{\partial \hat{y}}{\partial L}$

☐ K $\frac{\partial x_1}{\partial z}$

☐ B $\frac{\partial L}{\partial x_2}$

☐ G $\frac{\partial \hat{y}}{\partial z}$

☐ L $\frac{\partial x_2}{\partial z}$

☐ C $\frac{\partial L}{\partial \hat{y}}$

☐ H $\frac{\partial z}{\partial \hat{y}}$

☐ M $\frac{\partial z}{\partial w_1}$

☐ D $\frac{\partial x_1}{\partial L}$

☐ I $\frac{\partial z}{\partial x_1}$

☐ N $\frac{\partial w_1}{\partial z}$

☐ E $\frac{\partial x_2}{\partial L}$

☐ J $\frac{\partial z}{\partial x_2}$

☐ O None of the above

Q7.6 (1 point) Now, we want to use the partial derivative $\frac{\partial L}{\partial w_1}$ to update the value of w_1 to help our neural network predict better. Fill in the operator that best completes the update rule. α is the learning rate.

$w_1 \leftarrow w_1 \text{ — } \alpha \frac{\partial L}{\partial w_1}$

☐ A +

☐ B −

☐ C ×

☐ D ÷